Unit 4Relay

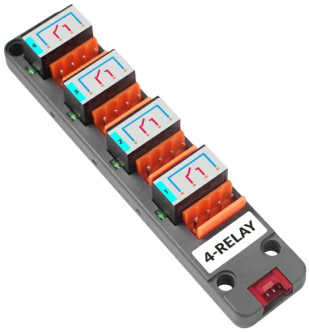
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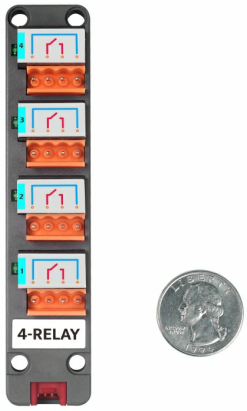












Description

Unit 4Relay is a high-voltage drive unit that integrates 4-channel relay control. It is based on the STM32F030F4P6 microcontroller and uses the I2C bus to independently control four mechanical relays, supporting high-voltage switching of AC 250V / DC 28V. The unit communicates via a Grove HY2.0-4P interface and features physical isolation, programmable status indicator LEDs, and overload capability. It also adopts a LEGO-compatible mounting hole design, making it easy to integrate with LEGO structures or secure using screws. This unit is suitable for high-voltage load switching scenarios such as smart home applications, industrial control, and power management of high-power equipment.

Features

- Channel Count: 4-channel independent relay control
- Load Capacity: AC 250V@10A or DC 28V@10A
- Status Indication: Independent programmable LED indicator for each channel
- Isolation Feature: Physical isolation between high-voltage and low-voltage sides
- 2 x LEGO-compatible mounting holes
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Unit 4Relay
- 1 x HY2.0-4P Grove connection cable (20cm)
- 4 x HT3.96-4P terminals

Applications

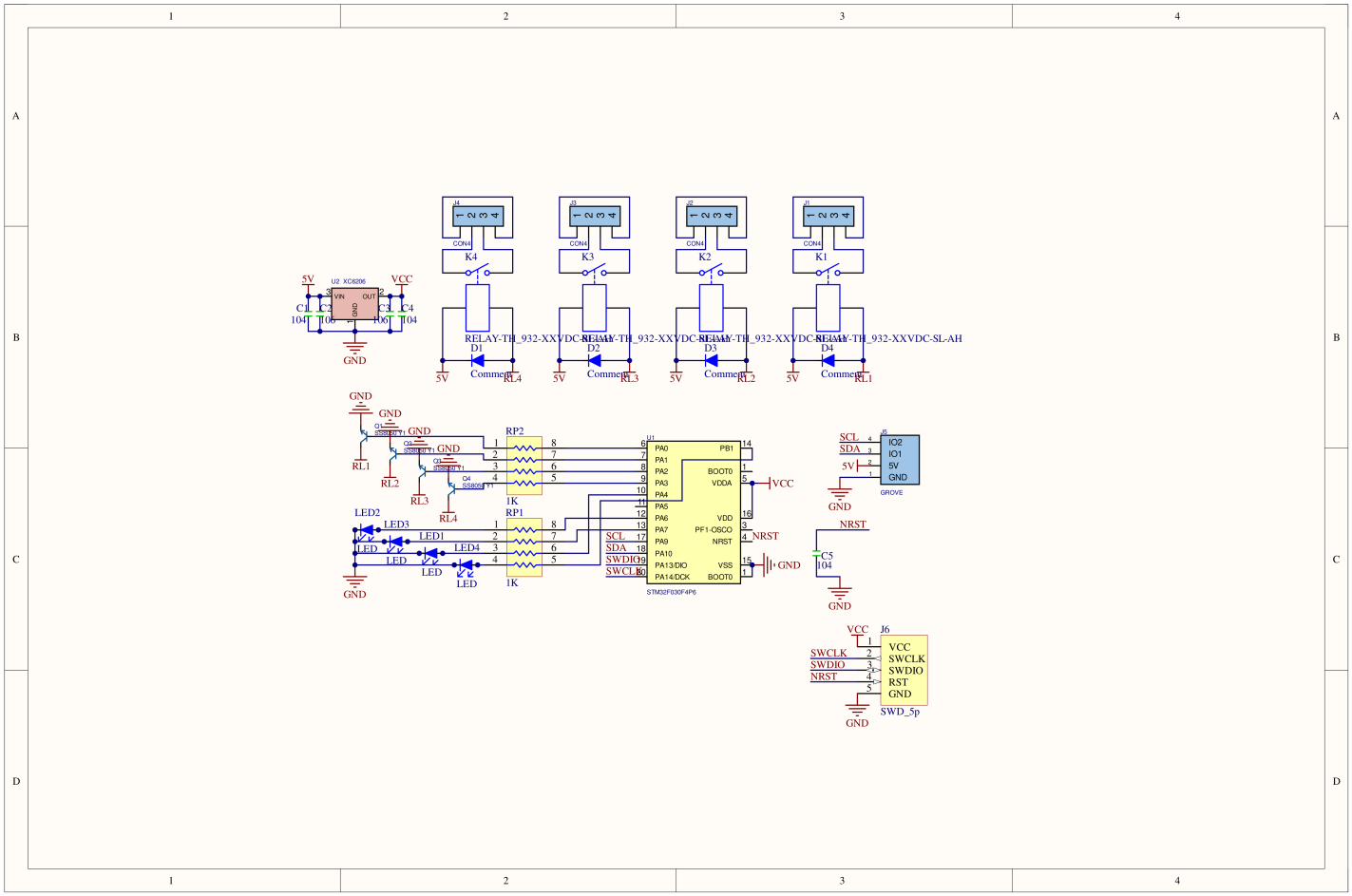
- DC signal switching
- Power on/off for digital devices

Specifications

Specification	Parameter
MCU	STM32F030F4P6
Max Input Voltage	AC 250V / DC 28V
Rated Current	10A
Max Instantaneous Current	16A
Communication Interface	I2C @0x26
Product Size	112.0 x 23.0 x 18.0mm
Product Weight	40.0g
Package Size	163.0 x 100.0 x 20.0mm
Gross Weight	84.0g

Schematics

- Unit 4Relay Schematics PDF

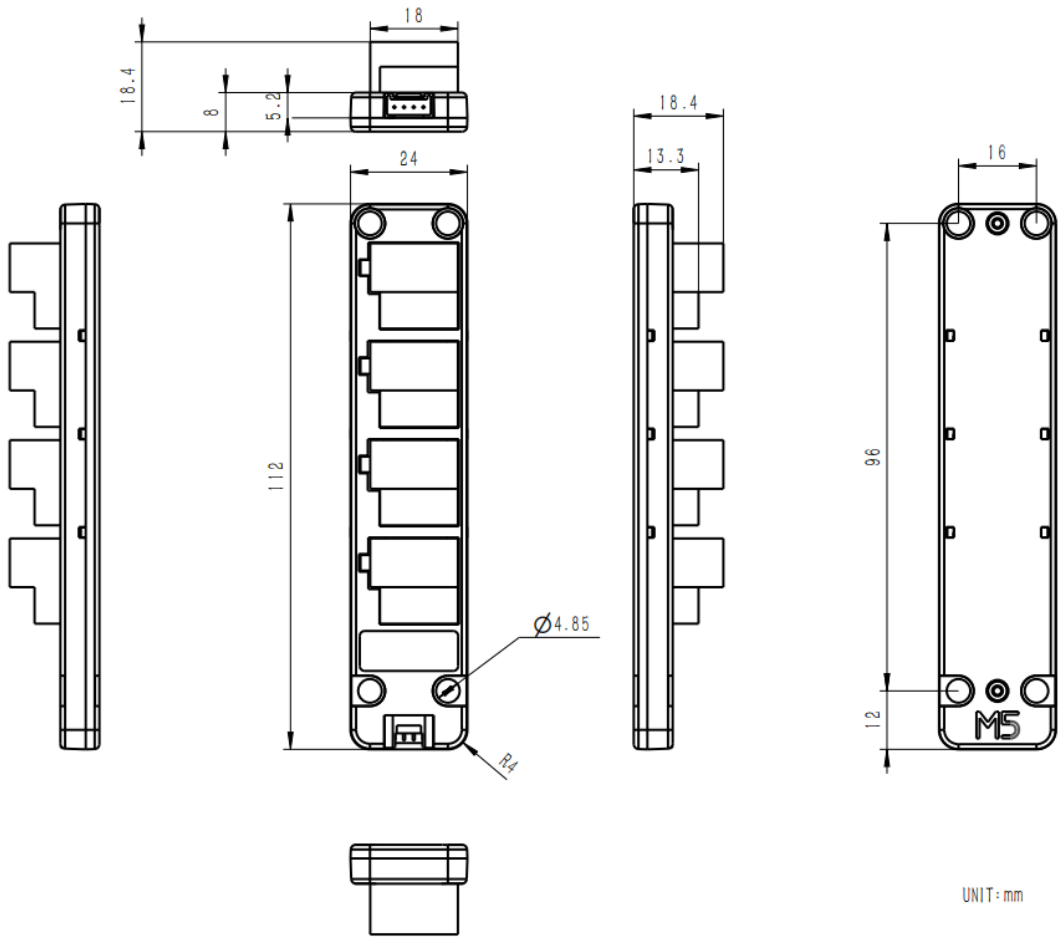


PinMap

Unit 4Relay

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit 4Relay Structure Files

Softwares

Arduino

- Unit 4Relay Arduino Tutorial

UiFlow1

- Unit 4Relay UiFlow1 Docs

UiFlow2

- Unit 4Relay UiFlow2 Docs

Home Assistant

- Unit 4Relay Home Assistant Integration

Protocol

- Protocol Type: I2C
- I2C Address: 0x26

Mode control Reg 0x10

Bit	Desc	R/W
7	R	R/W
6	R	R/W
5	R	R/W
4	R	R/W
3	R	R/W
2	R	R/W
1	R	R/W
0	LED Sync Mode 0:Manual 1:Auto	R/W

Working Mode Explanation

Manual: LED lights support independent control

Auto: LED follows Relay status changes, lights up when Relay is ON, turns off when OFF.

RELAY control Reg 0x11

Bit	Desc	R/W
7	LED1 / 1: ON 0:OFF	R/W
6	LED2 / 1: ON 0:OFF	R/W
5	LED3 / 1: ON 0:OFF	R/W
4	LED4 / 1: ON 0:OFF	R/W
3	RELAY1 / 1: ON 0:OFF	R/W
2	RELAY2 / 1: ON 0:OFF	R/W
1	RELAY3 / 1: ON 0:OFF	R/W
0	RELAY4 / 1: ON 0:OFF	R/W

EasyLoader

Easyloader	Download Link	Remarks
Unit 4Relay Example with M5Core	download	A key switches relay number, B key switches LED sync/async, C key controls all relays

Video

- Unit 4Relay Test with M5Core

4-RELAY_UNIT.mp4

- UiFlow2 Unit 4Relay